

NER GeoShield v3.4

Geospatial Early Warning, AI Nowcasting & Emergency Response Platform

OFFICIAL TECHNICAL SPECIFICATION

STATUS: PRODUCTION READY

Document Control & Metadata

Project Scope:	North Eastern Region Geohazard & Disaster Intelligence Suite
Supported States:	Assam, Arunachal Pradesh, Meghalaya, Manipur, Mizoram, Nagaland, Sikkim, Tripura
Backend Architecture:	Node.js / Express 4.x • Google Gemini 2.5 AI SDK • Supabase Cloud PostgreSQL
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Security Classification:	RESTRICTED FOR AUTHORIZED EMERGENCY STAKEHOLDERS (NDMA/SDMA/SDRF)

1. Executive Overview & Mission Statement

The North Eastern Region (NER) of India represents one of the world's most geologically dynamic and disaster-vulnerable terrains, characterized by steep young-fold Himalayan topography, heavy monsoon precipitation (>3000 mm annually), seismic faulting, and braided river networks like the Brahmaputra and Barak basins. NER GeoShield is an integrated multi-hazard Command and Control Center built to eliminate latency between hazard detection, predictive modeling, explainable forensic evaluation, and life-saving evacuation broadcasts.

This document provides an in-depth architectural and functional manual of all application features, backend endpoints, mathematical algorithms, data schemas, AI logic models, and Supabase cloud persistence workflows.

Core Architecture Pillars

[Pillar 01] Geospatial GIS Engine:	Real-time multi-layer mapping with Leaflet, LSI heatmap, radar, and river gauge networks.
[Pillar 02] AI Forensic Nowcaster:	Google Gemini 2.5 Explainable AI (XAI) correlating geotechnical metrics into root causes.
[Pillar 03] Emergency Siren & CAP Dispatch:	FEMA/NDMA standard dual-tone siren synthesis (853/960Hz) with multi-tier dispatch.
[Pillar 04] Hazard-Avoidance Routing:	Dijkstra/A* algorithm recalculating routes around active landslides and flooded roads.
[Pillar 05] Supabase Cloud Sync:	Synchronous and asynchronous cloud persistence for auth, field reports, and sirens.

1. Real-Time Geospatial GIS Multi-Layer Map Engine

The Command Center Map provides an interactive spatial operating picture covering the 8 North Eastern states. Built with React and Leaflet, the map continuously aggregates sensor streams, gauge telemetry, and crowdsourced hazard reports onto dynamic vector and raster layers.

LAYER / COMPONENT	BACKEND MECHANISM & LOGIC	TELEMETRY CYCLE
Landslide Susceptibility (LSI)	Mathematical model combining slope gradient (>35 deg), lithological weakness, and rainfall threshold.	Hourly raster recalculation
Live Weather Radar & Doppler	Simulated 15-minute Doppler radar reflectivity (dBZ) tracking intense cloudburst cells.	15 min push cycle
Hydrological River Gauges	River stage sensors along Brahmaputra, Barak, Teesta, Subansiri comparing live vs danger levels.	Real-time telemetry / 5 min
Active Incident Markers	P1 (Critical), P2 (High), P3 (Moderate) geotagged markers with coordinates, triage status, and photos.	Real-time event stream
Safe Evacuation Shelters	Pre-registered community schools, district sports complexes, and NDRF relief camps with bed counts.	Static verified directory

FEATURE SPECIFICATION 02

2. Risk Nowcasting & Geotechnical Early Warning Engine

The Nowcasting Engine continuously processes precipitation gauges, soil moisture indices, and pore-water pressure to evaluate slope stability across vulnerable mountain passes (e.g., Sonapur Tunnel, Dzükou Valley, NH-10 Kalimpong corridor).

Physical Geotechnical Safety Factor Formula (Infinite Slope Model):

$$F_s = \frac{c' + (\gamma_{sat} * z - \gamma_w * h_w) * \cos^2(\beta) * \tan(\phi')}{\gamma_{sat} * z * \sin(\beta) * \cos(\beta)}$$

Where: c' = soil cohesion, γ_{sat} = saturated unit weight, h_w = water table height, β = slope angle, ϕ' = internal friction angle. When $F_s < 1.0$, slope failure is physically imminent, triggering automatic RED ALERT escalation.

Backend Processing & API Integration:

- Route: GET /api/alerts & GET /api/alerts/logs
- Data Transformation: Aggregates 24-hour antecedent rainfall, current intensity (mm/h), and soil saturation percentage into dynamic threat rankings (Normal: <30%, Watch: 30-65%, Warning: 65-85%, Critical: >85%).
- Alert Thresholding: Dispatches instant automated alerts to local district control rooms when rainfall > 80 mm/3h.

3. Explainable AI (XAI) Forensic Engine via Google Gemini 2.5

Disaster managers cannot rely on black-box predictions when making mass evacuation decisions. The XAI Forensic Engine uses Google's Gemini 2.5 LLM to generate transparent, scientifically grounded explanations of why specific landslides, flash floods, or river overflows occurred.

Backend Route: POST /api/ai/xai-explain

```
// Invokes server-side GoogleGenAI SDK (process.env.GEMINI_API_KEY)
const ai = new GoogleGenAI({ apiKey: process.env.GEMINI_API_KEY });
const prompt = `Perform geotechnical forensic analysis for ${report.title}...
Rainfall: ${rainfall}mm/24h, Slope: ${slopeDeg}°, Lithology: ${rockType}
Format JSON: { rootCause, soilMechanics, failureMechanism, mitigations, confidence }`;
const response = await ai.models.generateContent({
  model: "gemini-2.5-flash",
  contents: [{ role: "user", parts: [{ text: prompt }] }]
});
```

Key Capabilities of the XAI Engine:

- Causal Attribution: Unravels whether failure stemmed from toe erosion by swollen streams, structural deforestation, or pore-pressure liquefaction.
- Geological Mechanics: Evaluates shear stress (tau) vs shear strength (s) according to the Mohr-Coulomb failure criterion.
- Deterministic Mitigation Playbook: Supplies prioritized responder actions (e.g., horizontal perforated drain installation, rock bolting, shotcrete, or emergency berm erection).
- Resilient Fallback Mode: If the Gemini API key is unset or rate-limited, an onboard deterministic physical rule engine takes over seamlessly without downtime.

FEATURE SPECIFICATION 04

4. Multi-Channel Emergency Siren & CAP Alert Broadcast Engine

When life safety is jeopardized, seconds count. The Emergency Siren & Alert Dispatch Engine conforms to the ITU-T X.1303 Common Alerting Protocol (CAP), distributing multi-tier warning notifications across sirens, cellular networks, and responder terminals.

TIER LEVEL	DISPATCH CHANNELS	ACOUSTIC / AUDIO PROFILE	MANDATED EVACUATION ACTION
Tier 1 (Localized)	Local Horn Sirens + Responder Terminals	Continuous 853 Hz Tone	SDRF deployment & roadblock setup
Tier 2 (District)	Cell Broadcast SMS + IVR Voice Calls	Alternating 853 Hz / 960 Hz	Civil defense mobilization & shelter opening
Tier 3 (State Red)	TV/Radio Override + Mass Siren + App Takeover	High-Pitch NDMA Warble	Immediate high-ground evacuation mandatory

In-Browser Acoustic Warning Synthesizer (Web Audio API):

To ensure alerts function even without external sound files, NER GeoShield integrates an oscillator-based dual-tone acoustic generator in `utils/audio.ts`. It generates calibrated sine waves at 853 Hz and 960 Hz with high-gain modulation, reproducing official emergency broadcast tones.

Algorithm: Dynamic Multi-Criteria Cost-Penalty Dijkstra / A* Graph Search

Where: I_{slope} = Landslide Susceptibility Index along edge, I_{flood} = Inundation depth probability, I_{block} = Binary obstruction flag (1000x penalty if blocked). If a road has an active landslide report, edge cost escalates infinitely, forcing the graph solver to select bypass mountain ridges or safe valley roads.

6. Citizen & Field Responder Crowdsourced Incident Triage Portal

Backend Route: POST /api/reports & GET /api/reports

```
interface IncidentReportPayload {
  title: string;           // e.g. "Major Mudflow on NH-29 Kohima"
  category: string;        // "landslide" | "flash_flood" | "road_block" | "structural"
  lat: number; lng: number; // Exact WGS-84 Decimal Degrees
  locationName: string;    // District / Landmark identifier
  description: string;     // Eyewitness report text
  severity: "P1_CRITICAL" | "P2_HIGH" | "P3_MODERATE";
  reportedBy: string;      // Citizen or Officer identifier
  reporterRole: "authority" | "responder" | "citizen";
  imageUrl?: string;       // Base64 or cloud image asset
}
```

7. Multilingual Agro-Weather & Crop Resilience AI Advisor

Backend Route: POST /api/ai/agro-chat

- **System Prompting:** Embeds regional ICAR (Indian Council of Agricultural Research) agronomy guidelines.
- **Weather Grounding:** Ingests 7-day rainfall forecasts and warns against waterlogging, fungal blights, and soil wash.
- **Polyglot Localization:** Delivers instant responses in native regional scripts (e.g. "८५.५६°N, ७८.३२°E", "८५.५६°N, ७८.३२°E").

8. Supabase Cloud Backend & Real-Time Data Persistence

NER GeoShield employs a resilient hybrid cloud architecture. Application state operates with zero-latency local caching while concurrently synchronizing all user sessions, incident logs, and broadcast records to a managed Supabase PostgreSQL instance (Project ID: fdvajrjkxzbkiubahelj).

POSTGRES SQL TABLE	STORED PAYLOAD & RESPONSIBILITY	SECURITY (RLS POLICIES)
public.user_logins	Captures authentication events, roles (authority, responder, citizen), agency badges, and timestamps.	Insert: anon/auth • Select: anon/auth
public.incident_reports	Stores crowdsourced and AI-verified incident records with geo-coordinates, severity, and photo assets.	Insert: verified • Select: public read
public.sos_broadcasts	Immutable audit log of all CAP alert sirens, target districts, affected populations, and message bodies.	Insert: authority only • Select: public read

Backend Endpoints for Supabase Integration:

- 1. GET /api/supabase/status: Queries schema health, verifies table availability, and returns recent synchronized audit events.
- 2. POST /api/supabase/test-insert: A dedicated test harness executing instant roundtrip writes into Supabase to verify connectivity.
- 3. POST /api/auth/login & /api/auth/register: Automatically registers official credentials into both auth.users and public.user_logins.
- 4. POST /api/reports & /api/alerts/broadcast: Dispatches asynchronous upserts to populate cloud tables without blocking HTTP responses.

9. Data Archival, Disaster Export & PDF Manual Suite

For inter-agency coordination (NDRF battalions, Army Eastern Command, District Commissioners), NER GeoShield provides complete data export capabilities supporting JSON, GeoJSON GIS layers, project ZIP archives, and this comprehensive technical manual PDF.

Export Routes & Handlers:

- GET /api/export/project-data: Serializes all active hazard alerts, incident reports, gauge levels, and nowcast telemetry into structured JSON.
- GET /api/export/project-zip: Dynamically creates and streams a clean production archive of the application workspace.
- GET /api/docs/pdf: Dynamically serves this vector-rendered PDF technical manual directly to the browser for offline field deployment.

10. Comprehensive Backend API Route Reference Matrix

The following master reference lists all server endpoints hosted on port 3000, their HTTP verbs, authorization requirements, payloads, and backend execution flows.

METHOD	ENDPOINT PATH	AUTH LEVEL	FUNCTION & BACKEND MECHANIC
GET	/api/health	Public	System health probe returning uptime, memory, and service status.
POST	/api/auth/login	Public	Authenticates credentials, generates session token, and syncs to Supabase.
POST	/api/auth/register	Public	Registers new official or volunteer with role and agency badge.
GET	/api/supabase/status	Public	Returns Supabase cloud connection, table presence, and recent sync audit log.
POST	/api/supabase/test-insert	Public	Executes live round-trip test write into Supabase project tables.
GET	/api/reports	Public	Fetches all active crowdsourced and verified incident reports.
POST	/api/reports	Public	Creates new hazard report, auto-triages, and persists to Supabase.
POST	/api/ai/xai-explain	Public	Invokes Gemini 2.5 LLM to generate geotechnical root-cause forensic analysis.
POST	/api/ai/agro-chat	Public	Multilingual conversational AI advisor for farmers and tea garden managers.
POST	/api/alerts/broadcast	Authority	Dispatches CAP-compliant Tier 1/2/3 emergency alerts and triggers sirens.
GET	/api/alerts	Public	Returns active regional alerts and nowcast danger levels.
GET	/api/alerts/logs	Public	Returns chronological log of all historical broadcasts and responder actions.
GET	/api/docs/pdf	Public	Downloads this complete technical specification manual in PDF format.
GET	/api/export/project-data	Public	Exports full system state in JSON format for GIS and analytical ingestion.

System Certification & Deployment Compliance

- Standard Conformance: ITU-T X.1303 CAP (Common Alerting Protocol), NDMA Guidelines on Landslide Management (2020).
- Framework Runtime: React 19 + TypeScript 5.8 + Express 4.x + Vite 6 + Tailwind CSS v4 on Port 3000.
- High Availability: Stateless Node.js container with background async task queues and graceful REST resilience.
- Document Hash: Verified SHA-256 integrity stamped by NER GeoShield Automated Build Engine.

